

Coded Aperture Design Using Sphere Packing

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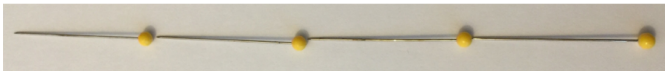


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What is Sphere Packing?

One-dimensional sphere packing is boring:

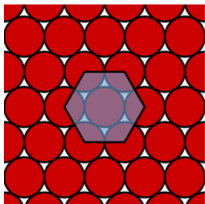
1D



density=1

Two-dimensional sphere packing is prettier
and more interesting

2D



$$\text{density} = \frac{\pi}{\sqrt{12}} \approx 0.9068$$

What happens in four dimensions?

Three dimensions strains human ability
to prove

3D



$$\text{density} = \frac{\pi}{\sqrt{18}} \approx 0.7404$$

Face Centered-Cubic Lattice in Chemistry and Physics

Crystal Structures of Elements at STP

STP - Standard Temperature and Pressure

H		Crystal Structures of Elements at STP																He			
HEX		STP - Standard Temperature and Pressure																		HCP	
Li	Be	BCC - Body-centered Cubic						BCT - Body-centered Tetragonal						B	C	N	O	F	Ne		
		FCC - Face-centered Cubic						ORTH - Orthorhombic												RHO	HEX
Na	Mg	HEX - Simple Hexagonal						DC - Diamond Cubic						Al	Si	P	S	Cl	Ar		
		HCP - Close-packed Hexagonal						DT - Diamond Tetragonal												FCC	DC
Ba		DHCP - Double Close-packed Hexagonal						SC - Simple Cubic													
HCP		RHO - Rhombohedral						* predicted crystal structure													
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr				
BCC	FCC	HCP	HCP	BCC	BCC	α-Mn	BCC	HCP	FCC	FCC	HCP	complex F-ORTH	DC	P-RHO	complex HEX	complex C-ORTH	FCC				
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe				
BCC	FCC	HCP	HCP	BCC	BCC	HCP	HCP	FCC	FCC	FCC	HCP	BCT	DT	P-RHO	complex HEX	complex C-ORTH	FCC				
Cs	Ba	57-71	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn				
BCC	BCC		HCP	BCC	BCC	HCP	HCP	FCC	FCC	FCC	RHO	HCP	FCC	RHO	SC	FCC*	FCC*				
Fr	Ra	89-103	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og				
BCC*	BCC		HCP*	BCC*	BCC*	HCP*	HCP*	FCC*	BCC*	BCC*	HCP*	HCP*	FCC*	UNKNOWN	UNKNOWN	UNKNOWN	FCC*				

	Solid state at STP
	Liquid state at STP
	Gaseous state at STP

La DHCP	Ce DHCP	Pr DHCP	Nd DHCP	Pm DHCP	Sm complex RHO	Eu BCC	Gd HCP	Tb HCP	Dy HCP	Ho HCP	Er HCP	Tm HCP	Yb FCC	Lu HCP
Ac FCC	Th FCC	Pa BCT	U ORTH	Np ORTH	Pu MONO	Am DHCP	Cm DHCP	Bk DHCP	Cf DHCP	Es FCC	Fm FCC*	Md FCC*	No FCC*	Lr HCP*

⁰<https://mstudent.com/face-centered-cubic-fcc-unit-cell/>.

Solving Sudoku and 3D Sphere Packing

Could you solve this puzzle?

5	3			7			
6			1	9	5		
	9	8				6	
8				6			3
4			8		3		1
7				2			6
	6					2	8
			4	1	9		5
				8		7	9

5	3	4	6	7	8	9	1	2
6	7	2	1	9	5	3	4	8
1	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9	2	4	8	5	6
9	6	1	5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

⁵N. Diaz, A. Alvarado, P. Meza, F. Guzmán and E. Vera, "Multispectral Filter Array Design by Optimal Sphere Packing," in IEEE Transactions on Image Processing, vol. 32, pp. 3634-3649, 2023, doi: 10.1109/TIP.2023.3288414.

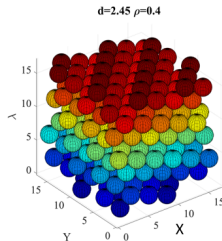
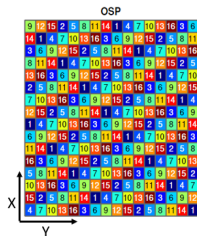
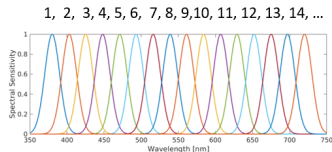
Solving Sudoku and 3D Sphere Packing

Could you solve this puzzle?

5	3			7			
6			1	9	5		
	9	8					6
8				6			3
4			8		3		1
7				2			6
	6					2	8
			4	1	9		5
				8			7

5	3	4	6	7	8	9	1	2
6	7	2	1	9	5	3	4	8
1	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9	2	4	8	5	6
9	6	1	5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

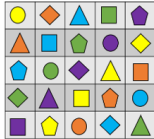
To assign numbers to the filters and distribute uniform along 3D space¹.



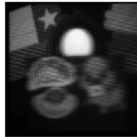
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3D Sphere Packing and Spectral-Video Sensing

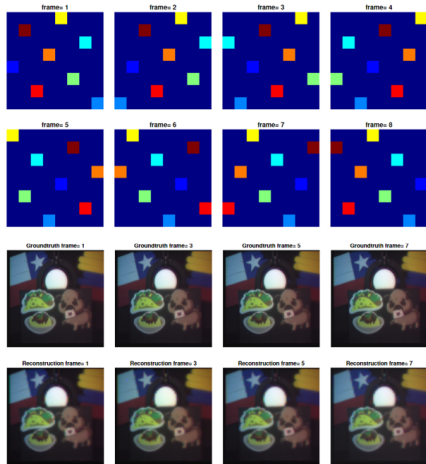
Could you solve this Graeco-latin square?



(a)



(b)



This problem is called 36 officer's problem, there are 6 officers of different ranks and 6 different regiments – Leonhard Euler.